ℓ^1-Native Cohomology of Defect Fields: Quotient Norms, Dual Circulations, and Hodge Inflation

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> **Status:** core quotient-obstruction paper
> **Carrier:** finite connected graphs without 2-cells; real scalar 1-cochains with a declared ℓ^1 cochain norm
> **Observer:** assumes the replica-extensive, edge-additive observer class that selects ℓ^1 on C^1; does not re-prove norm selection
> **Main claims:** quotient norm Φ on $Q^1 = C^1/\operatorname{im}(d_0)$; finite primal-dual certificate by bounded divergence-free circulations; explicit ℓ^2 Hodge inflation example on C_n
> **Non-claims:** no universal ℓ^1, no ℓ^1 Hodge theorem, no universal inflation bound, no new optimization algorithm, no lift authority from Φ alone, no quantum extension theorem
> **Dependency:** specializes the General Theory normed-cokernel construction to finite graphs; logical inputs include Paper 000 observer discipline and the declared ℓ^1-selection papers

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Classification of Results

This paper separates rigorous mathematics from original constructions and speculative analogies. Each result is annotated with its epistemic layer:

Layer	Status	What it contains
A: Theorems	Established mathematics	Standard results from discrete algebraic topology and discrete ℓ^2 Hodge theory on finite graphs without 2-cells [1, 7, 8].
B: Paper-specific organization	Conditional finite construction	The notation Φ for the ℓ^1 quotient norm on graph cohomology, the finite primal-dual certificate presentation, and the explicit C_n inflation calculation. LP duality and quotient norms themselves are standard.
C: Analogy	Structural analogy	The parallel to anisotropic total variation (graph-TV) denoising in image processing [2, 13]. No convergence or algorithmic performance theorem is claimed.

Abstract

The General Theory treats exact repairs as $B^1 = \mathrm{im}(d_0)$, quotient obstruction as $Q^1 = C^1/B^1$, and a declared finite gauge as inducing a normed cokernel (Q^1, Φ). This paper specializes that construction to finite graphs without 2-cells. It assumes an ℓ^1 norm on C^1 already selected by a declared edge-additive observer class and asks how to measure the irreducible component of a defect field under that chosen cochain geometry.

On finite graphs, classical discrete Hodge theory provides an ℓ^2-orthogonal decomposition separating exact and harmonic components. However, the ℓ^2-orthogonal Hodge decomposition is not the same object as the ℓ^1 quotient minimization: its component bookkeeping cost can differ from, and in the cycle example below exceed, the sparse representative's ℓ^1 cost.

We study the functional

$$\Phi([\delta]) = \min_{f \in C^0} \|\delta - d_0 f\|_1,$$

which is the standard ℓ^1 quotient norm on $Q^1 = C^1/\mathrm{im}(d_0)$ (the minimum is attained; see Proposition 7.1). While this optimization problem is classical (appearing in total variation minimization and minimum homologous chain problems [12]), the point here is narrower: **given the ℓ^1 norm on C^1, this is the quotient norm induced by the standard quotient construction** (Theorem 7.2, Remark 7.7).

We further relate Φ to discrete Hodge decomposition, prove that the ℓ^2 Hodge component bookkeeping cost can inflate relative to the sparse representative for a specific single-edge construction on cycle graphs C_n, with ratio $3 - 2/n$, and characterize Φ via linear programming duality as the dual norm of bounded divergence-free circulations. This provides a unified interpretation connecting cohomology, ℓ^1 optimization, and network flows, without introducing new optimization machinery.

Core Theorem Statement

Theorem 0.1 (Finite Graph ℓ^1 Quotient-Dual Obstruction). Let $G = (V, E)$ be a finite connected graph with a fixed orientation, incidence matrix B, and cochain spaces $C^0 = \mathbb{R}^V$, $C^1 = \mathbb{R}^E$. Assume C^1 is equipped with the declared ℓ^1 norm. For each $\delta \in C^1$, define

$$\Phi([\delta]) = \inf_{f \in C^0} \|\delta - Bf\|_1.$$

Then:

1. The infimum is attained.
2. Φ is the standard quotient norm on $C^1/\mathrm{im}(B)$.

3. The dual formula holds:

$$\Phi([\delta]) = \max_{\phi \in C^1} \{\langle \phi, \delta \rangle : B^T \phi = 0,\ \|\phi\|_\infty \leq 1\}.$$

4. Every primal minimizer f^* gives a decomposition $\delta = Bf^* + r^*$ with $\|r^*\|_1 = \Phi([\delta])$, certified by an optimal bounded circulation ϕ^*.

Because graphs here have no 2-cells, $d_1 = 0$ and every nonzero quotient class is closed. In the General Theory trichotomy, $\Phi([\delta]) = 0$ is exact repair, while $\Phi([\delta]) > 0$ is a stabilized survivor rather than an open obstruction. This theorem is conditional on the declared ℓ^1 norm on C^1; it does not assert that ℓ^1 is the only possible cochain norm.

Methodological Note: ℓ^1-Native Cohomology and Duality

The ℓ^2 Structure is an Optional Coordinate Scaffold; The Quotient Measurement is Convex-Geometric

1. The Category Shift: From Orthogonality to Duality

This version fixes the role separation. The ℓ^2 Hodge decomposition is a valid finite-dimensional orthogonal coordinate system. It is not, however, the structure that defines the ℓ^1 quotient measurement. In some classes the harmonic representative may be ℓ^1 optimal; in others it need not be. The point here is that Φ is defined by quotient minimization, not by orthogonal projection. The ℓ^1 construction below supplies an **optimization structure on equivalence classes**, not a replacement theorem for Hodge theory.

Key distinction: - **Classical (^{2}):** Subspaces, inner products, orthogonal projections, and dense harmonic representatives. - **Intrinsic (^{1}):** Polyhedral structure, convex duality, LP minimizers, and favoring sparse representatives.

2. The ℓ^1-Native Decomposition

The canonical measurement structure for the assumed ℓ^1 geometry is not orthogonal splitting but **Linear Programming Duality** (Section 10). For any δ, we possess a primal-dual pair derived from the quotient norm Φ: - **Primal Splitting:** $\delta = d_0 f^* + r^*$, dividing the field into a removable gauge artifact $d_0 f^*$ and an ℓ^1-minimal irreducible topological obstruction r^*. - **Dual Witness:** Certified by a bounded divergence-free circulation ϕ^* ($B^T\phi^* = 0$, $\|\phi^*\|_\infty \leq 1$) that explicitly pairs with r^*.

This structure serves a different role from the ℓ^2 Hodge decomposition. Hodge theory gives a linear orthogonal coordinate system; the primal-dual quotient construction gives the ℓ^1-minimal representative and a circulation certificate. The inflation ratio $3 - 2/n$ for the single-edge cycle construction in Section 6

quantifies an explicit mismatch between ℓ^2 Hodge component bookkeeping and the sparse representative's ℓ^1 cost. No universal inflation bound is claimed.

3. Affine Constraint Structure

For $\omega \in C^1$, the feasible residuals are:

$$h_1 \in \omega - \mathrm{im}(d_0).$$

This affine structure determines the space over which Φ is minimized.

The affine constraint enforces feasibility of h_1; the norm determines which feasible residual is selected; and the dual witness certifies optimality and non-exactness when the optimum is positive.

Thus:

- topology $\to$ feasible set
- norm $\to$ selection rule

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1. Introduction

The preceding papers in this series establish, respectively, the observer discipline for quotient obstruction [18], M-compatible gauge classification for Banach presheaves [15], coordinate-wise ℓ^1 selection on finite graph cochains [16], and projection-induced coboundary defects in factorized dynamics [14]. The numbering of the preprint series reflects publication order, not logical dependency; the logical inputs are the observer framework, the declared ℓ^1 cochain geometry, and the projection-defect source model.

A natural next question arises: given a defect field $\delta \in C^1(G)$, how should one decompose it into removable (exact) and irremovable (cohomological) components? The classical tool is the discrete Hodge decomposition, which operates in ℓ^2. But given the ℓ^1 structure established in [14–16] as the baseline diagnostic geometry, a tension emerges.

Throughout, we assume an ℓ^1 geometry on C^1 as selected in the replica-extensive observer setting of [15, 16] and used by the projection-defect model [14]; this paper studies its consequences. The central problem addressed here is therefore not to choose a convenient norm, but to determine the quotient norm induced by that ℓ^1 cochain metric.

Classical Hodge decomposition does not generally hold in L^1 Banach spaces, as they lack the inner-product structure necessary to define orthogonal projections. We do not construct an ℓ^1 Hodge decomposition. Instead, we present the **ℓ^1-Native Decomposition** driven by convex duality and linear

programming. The ℓ^2 Hodge frame remains useful as an algebraic coordinate system; it is simply not the structure that minimizes ℓ^1 quotient cost.

We demonstrate that the ℓ^1 quotient norm determines the induced quotient magnitude and gives an optimization-based splitting into gauge-removable and irreducible components. The resulting **Cohomological ℓ^1 Norm** is the quotient norm induced by ℓ^1 on C^1: it evaluates a class not by orthogonal projection, but by infimal ℓ^1 gauge cost and maximum dual flow pairing.

Scope and Relation to Existing Work. We emphasize that the optimization problem $\min_f \|\delta - d_0 f\|_1$ is not new. It appears in several established contexts, including anisotropic total variation on graphs [2, 13], ℓ^1 regression over incidence matrices [4], the minimum homologous chain problem [12], and is related to Gromov's bounded cohomology [11] in the infinite-dimensional setting. Compressed sensing and basis pursuit [5] exploit the same ℓ^1 sparsity structure, and shortest cycle basis algorithms [6] address the underlying combinatorial structure. The contribution of this paper is not to introduce a new optimization framework, but to identify this functional as the standard quotient norm induced by the assumed ℓ^1 structure on C^1 and to organize its primal-dual certificates within a cohomological framework.

Recall (declared ℓ^1 observer class). The ℓ^1 coboundary norm $I(s) = \sum_e \|\delta_e(s)\|$ is selected in [15, 16] under explicit observer assumptions: edge decomposability, replica extensivity, symmetry/naturality constraints, and normalization. In the presheaf language of [15], the coboundary is $\delta_e(s) = r_{b,a}(s(b)) - s(a)$. When the Banach presheaf F is the constant sheaf $\mathbb{R}$ on a graph G, this reduces to $(d_0 f)(e) = f(v) - f(u)$, the total norm $I(s)$ becomes $\|d_0 f\|_1$, and the quotient norm $\Phi([\delta]) = \min_f I(\delta - d_0 f)$ measures the irreducible cost after gauge optimization. This paper develops the structure of Φ after that observer class has been declared.

Notation bridge. The following dictionary connects the presheaf notation of the earlier obstruction papers to the graph notation used in the remainder of this paper:

Presheaf notation	Graph notation (this paper)
Poset P, covering relations $E_{\mathrm{cov}}(P)$	Graph $G = (V, E)$
Banach presheaf F, stalks $F(x)$, restrictions $r_{b,a}$	Constant sheaf $\mathbb{R}$, identity restrictions
Coboundary $\delta_e(s) = r_{b,a}(s(b)) - s(a)$	Discrete derivative $(d_0 f)(e) = f(v) - f(u)$
ℓ^1 coboundary norm $I(s) = \sum_e \|\delta_e(s)\|$	$\|d_0 f\|_1 = \sum_e \lvert(d_0 f)_e\rvert$

Conventions. Throughout this paper, $G = (V, E)$ denotes a finite, connected, undirected graph. Each edge is assigned an arbitrary but fixed orientation, used only for algebraic bookkeeping; all results are independent of this choice. We

work over the scalar field $\mathbb{R}$. Norms are denoted $\|\cdot\|$ with appropriate subscripts (ℓ^1, ℓ^2, ℓ^∞); absolute values of scalars are denoted $|\cdot|$. The extension to vector-valued defects $\delta \in C^1(G, \mathbb{R}^k)$ is observer-class dependent and is discussed in Section 11.

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2. Gauge Equivalence (Layer A/B)

Before constructing the decomposition machinery, we introduce the central gauge-theoretic modeling notion that motivates the entire analysis.

A defect field $\delta \in C^1(G)$ records the local projection error across each edge. However, the value of δ on a single edge is not gauge-invariant; it depends on the choice of local reference state at each vertex. A **gauge transformation** is a reassignment of these local reference states, parameterized by a vertex function $f \in C^0(G) = \mathbb{R}^{|V|}$. We use "gauge equivalence" in the sense of equivalence under reassignment of local reference frames, analogous to the gauge freedom of electromagnetic potentials $A \mapsto A + d\Lambda$.

Definition 2.1 (Gauge Equivalence). Two defect fields $\delta, \delta' \in C^1(G)$ are **gauge equivalent** if their difference is an exact gradient:

$$\delta \sim \delta' \iff \delta - \delta' \in \operatorname{im}(d_0)$$

where $d_0 : C^0 \to C^1$ is the discrete exterior derivative defined in Section 3. The equivalence class $[\delta] = \delta + \operatorname{im}(d_0)$ is the **gauge class** of δ.

The observer-invariant quantity in this model is not the representative δ itself but its gauge class $[\delta]$. The natural gauge-invariant magnitude of an obstruction is therefore:

$$\operatorname{Cost}(\delta) = \min_{f \in C^0} \|\delta - d_0 f\|_1$$

This is precisely the ℓ^1 quotient norm on $C^1/\operatorname{im}(d_0)$. It represents the minimal ℓ^1 cost achievable over all gauge-equivalent representatives. We will show that this quantity defines a norm on the first cohomology group $H^1(G, \mathbb{R})$ and admits a clean dual formulation connecting it to the topological cycle structure of G.

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3. The Discrete de Rham Complex (Layer A)

Let $C^0 = \mathbb{R}^{|V|}$ denote the space of 0**-cochains** (vertex functions) and $C^1 = \mathbb{R}^{|E|}$ denote the space of 1**-cochains** (edge fields). Each edge e carries its fixed orientation $e = (u, v)$.

Definition 3.1 (Discrete Exterior Derivative). The operator $d_0 : C^0 \to C^1$ is the graph difference operator:

$$(d_0 f)(e_{uv}) = f(v) - f(u)$$

Its matrix representation is the signed incidence matrix $B \in \mathbb{R}^{|E|\times|V|}$, with $B_{e,v} = +1$, $B_{e,u} = -1$ for each oriented edge $e = (u, v)$.

Definition 3.2 (Transpose Divergence). Let B^T be the transpose of the signed incidence matrix. The divergence operator is the algebraic map $B^T : C^1 \to C^0$:

$$\langle d_0 f, \omega\rangle_{\ell^2} = \langle f, B^T\omega\rangle_{\ell^2}$$

When standard ℓ^2 inner products are chosen, B^T is the adjoint usually denoted d_0^*. The duality theory below uses only the transpose equation $B^T\phi = 0$; it does not require an ℓ^2 metric. Explicitly, the **divergence** of a 1-cochain ω at a vertex u is the net signed flow:

$$(d_0^*\omega)(u) = \sum_{e \text{ incoming at } u} \omega(e) - \sum_{e \text{ outgoing from } u} \omega(e)$$

The discrete de Rham complex for a graph without 2-cells is:

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{0} 0$$

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4. The Hodge Laplacian and Harmonic Cochains (Layer A)

Following Eckmann [1], Horak–Jost [7], and Lim [8], the full 1-form Laplacian on a simplicial complex is:

$$\Delta_1 = d_0\, d_0^* + d_1^*\, d_1$$

Remark 4.1 (Convention and Missing 2-Cells). For a graph (a 1-dimensional simplicial complex), there are no 2-cells. Consequently $d_1 \equiv 0$, and the Laplacian collapses to:

$$\Delta_1 = d_0\, d_0^* \quad \text{(graphs without 2-cells)}$$

Definition 4.2. A 1-cochain $\omega \in C^1$ is **harmonic** if $\Delta_1\omega = 0$: $\mathcal{H}^1(G) = \ker \Delta_1$.

Proposition 4.3. *For a graph without* 2*-cells,* $\ker \Delta_1 = \ker d_0^*$.

Proof. $\ker d_0^* \subseteq \ker \Delta_1$ is immediate. Conversely, $\Delta_1\omega = 0$ implies $0 = \langle d_0 d_0^*\omega, \omega\rangle_{\ell^2} = \|d_0^*\omega\|_{\ell^2}^2$, so $d_0^*\omega = 0$.

Proposition 4.4. *If G is connected,* $\dim \mathcal{H}^1(G) = \beta_1(G) = |E| - |V| + 1$.

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5. The Discrete Hodge Decomposition (Layer A)

On a general continuous manifold, the Hodge theorem splits any 1-form into three orthogonal components: $\omega = d\alpha + \delta\beta + h$ (exact, coexact, and harmonic). However, because our state space is a finite graph with no 2-cells, $d_1 = 0$, its adjoint $d_1^* = 0$, and the coexact term vanishes entirely. The decomposition reduces to exactly two functional pieces.

Theorem 5.1 (Bipartite Hodge Decomposition). *Every $\delta \in C^1(G)$ decomposes algebraically uniquely as:*

$$\delta = \delta_{\text{ex}} + \delta_{\text{harm}}, \quad \delta_{\text{ex}} \in \text{im}(d_0), \quad \delta_{\text{harm}} \in \ker d_0^*, \quad \delta_{\text{ex}} \perp_{\ell^2} \delta_{\text{harm}}$$

Proof. By the orthogonal complement theorem in finite dimensions, $C^1 = \text{im}(d_0) \oplus (\text{im}(d_0))^\perp$. Since $\langle d_0 f, h\rangle_{\ell^2} = \langle f, d_0^* h\rangle_{\ell^2}$ for all f, h, we have $(\text{im}(d_0))^\perp = \ker(d_0^*)$. Therefore $C^1 = \text{im}(d_0) \oplus \ker(d_0^*)$ with the two summands ℓ^2-orthogonal.

Remark 5.2 (Finite-Dimensional Availability). This theorem explicitly relies on the finite-dimensionality of the graph. Infinite-dimensional L^1 spaces lack the inner-product structure necessary to define orthogonal projections, making L^1 Hodge theory structurally different from the L^2 case (cf. Kakutani's characterization of abstract L-spaces [17]). We do not attempt an ℓ^1 Hodge decomposition. Instead, we use the ℓ^2 Hodge decomposition as an algebraic coordinate system on $C^1(G)$ — guaranteed to exist in finite dimensions — and then evaluate its components under the ℓ^1 norm. The resulting mismatch reflects the incompatibility between orthogonal projection in ℓ^2 and sparsity structure in ℓ^1.

Remark 5.3 (Topological Interpretation). Under this framework, $\text{im}(d_0)$ corresponds to potential difference relationships (gauge transformations), while $\ker d_0^*$ captures network circulations. The harmonic component δ_{harm} is the gauge-invariant core, unchanged under $\delta \mapsto \delta - d_0 g$. For trees ($\beta_1 = 0$), all defects are exact and trivially gauge-removable. For graphs with loops ($\beta_1 > 0$), any non-zero harmonic component constitutes an irreducible topological obstruction.

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6. The ℓ^1/ℓ^2 Tension and Conditional Minimality (Layer A/B)

Remark 6.1 (Layer A). $\|x + y\|_1 \leq \|x\|_1 + \|y\|_1$, with equality iff $x_e y_e \geq 0$ for all e. This is the standard triangle inequality.

Remark 6.2 (Layer A). Applying the triangle inequality to the Hodge decomposition: $\|\delta\|_1 \leq \|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1$, with equality iff there is edgewise sign compatibility between the exact and harmonic components.

Remark 6.3. Generic sign conflicts make the inequality strict. Thus, the ℓ^2 algebraic Hodge projection operator generically yields strictly larger ℓ^1 component sums except in sign-compatible cases.

The cycle examples below show that an ℓ^2 coordinate representative can be a poor proxy for ℓ^1 quotient cost. They do not show that ℓ^2 is globally invalid, nor do they give a worst-case bound over all graphs.

Proposition 6.4 (Hodge Component-Cost Inflation Need Not Equal Quotient Cost) (Layer B). Let $\delta = \delta_{\text{ex}} + \delta_{\text{harm}}$ be the finite-dimensional ℓ^2 Hodge decomposition on a graph. The quantity

$$\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1$$

is not, in general, equal to the quotient norm $\Phi([\delta])$, nor is it a canonical ℓ^1 decomposition cost. It is the ℓ^1-cost of the two ℓ^2-orthogonal bookkeeping components.

In particular, for a single-edge defect on C_n, this component-cost ratio is

$$\frac{\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1}{\|\delta\|_1} = 3 - \frac{2}{n}.$$

This does not imply that δ_{harm} fails to minimize ℓ^1 cost in its cohomology class. In fact, for this specific cycle example, $\|\delta_{\text{harm}}\|_1 = \Phi([\delta]) = 1$. The example shows that ℓ^2 Hodge splitting can inflate component bookkeeping cost relative to the sparse representative, not that the harmonic representative is always ℓ^1-suboptimal.

6.1 Triangle Graph (Layer B)

Example 6.5. On C_3 with $\beta_1 = 1$, cycle vector $c = (1,1,1)^T$. A sparse local defect $\delta = (1,0,0)^T$ resolves via ℓ^2 projection to: - $\delta_{\text{harm}} = \frac{1}{3}(1,1,1)^T$, $\delta_{\text{ex}} = \frac{1}{3}(2,-1,-1)^T$ - $\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1 = \frac{4}{3} + 1 = \frac{7}{3} > 1 = \|\delta\|_1$ - Inflation ratio: $7/3 \approx 2.33$

6.2 Square Cycle (Layer B)

Example 6.6. On C_4 with $\delta = (1,0,0,0)^T$: - $\delta_{\text{harm}} = \frac{1}{4}(1,1,1,1)^T$, $\delta_{\text{ex}} = \frac{1}{4}(3,-1,-1,-1)^T$ - $\|\delta_{\text{harm}}\|_1 = 1$, $\|\delta_{\text{ex}}\|_1 = \frac{3}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{6}{4} = \frac{3}{2}$ - $\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1 = \frac{5}{2} > 1 = \|\delta\|_1$ - Inflation ratio: $5/2 = 2.5$

6.3 Inflation Ratio on Cycle Graphs (Layer B)

Proposition 6.7 (ℓ^2 Inflation Ratio on C_n). *For a single-edge defect $\delta = \mathbf{e}_1 = (1,0,\ldots,0)^T$ on the cycle graph C_n ($n \geq 3$), the ℓ^2 Hodge decomposition inflation ratio is:*

$$\frac{\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1}{\|\delta\|_1} = 3 - \frac{2}{n}$$

Proof. The harmonic space of C_n is $\ker(d_0^*) = \mathrm{span}(\mathbf{1})$, where $\mathbf{1} = (1, 1, \ldots, 1)^T$. The ℓ^2-orthogonal projection onto $\ker(d_0^*)$ gives:

$$\delta_{\text{harm}} = \frac{\langle \delta, \mathbf{1} \rangle}{\langle \mathbf{1}, \mathbf{1} \rangle} \mathbf{1} = \frac{1}{n} \mathbf{1}$$

$$\delta_{\text{ex}} = \delta - \delta_{\text{harm}} = \left(\frac{n-1}{n}, \frac{-1}{n}, \ldots, \frac{-1}{n} \right)$$

Computing the ℓ^1 norms:

$$\|\delta_{\text{harm}}\|_1 = n \cdot \frac{1}{n} = 1$$

$$\|\delta_{\text{ex}}\|_1 = \frac{n-1}{n} + (n-1) \cdot \frac{1}{n} = \frac{2(n-1)}{n}$$

The inflation ratio is:

$$\frac{\|\delta_{\text{ex}}\|_1 + \|\delta_{\text{harm}}\|_1}{\|\delta\|_1} = \frac{\frac{2(n-1)}{n} + 1}{1} = \frac{2n - 2 + n}{n} = \frac{3n-2}{n} = 3 - \frac{2}{n}$$

This grows monotonically with n and approaches 3 as $n \to \infty$. This inflation reflects the incompatibility between ℓ^2 projection and ℓ^1 geometry for this specific construction, rather than an intrinsic property of every defect field.
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Scope of Proposition 6.7. The formula applies to single-edge defects on cycle graphs C_n. General graphs may exhibit different behavior depending on cycle basis geometry, edge weights, and graph spectrum. No universal inflation bound is claimed here.

Remark 6.8 (Crosspolytope Bias). There is a geometric reason why the ℓ^2 harmonic projection can increase ℓ^1 magnitude. The mismatch stems from the difference between Euclidean balls (ℓ^2) and crosspolytopes (ℓ^1) in high dimensions. The ℓ^2 orthogonal projection minimizes Euclidean distance and often diffuses mass across coordinates uniformly (e.g., $1/3, 1/3, 1/3$). The ℓ^1 cost, dictated by the crosspolytope geometry, can privilege sparse, extreme configurations (corners). Proposition 6.7 demonstrates this on cycle graphs; it is not a theorem about all graphs.

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7. The Cohomological ℓ^1 Norm (Layer B)

This section develops the **Cohomological ℓ^1 Norm**, a term we adopt for the ℓ^1 quotient norm on $C^1/\operatorname{im}(d_0)$. The quotient norm construction is standard in functional analysis [4, 9]; the ℓ^1 quotient norm on homology classes appears in the bounded cohomology literature [11] and the minimum homologous chain

problem [12]. The contribution here is not the optimization framework itself, but the identification of Φ as the quotient norm induced by the assumed ℓ^1 geometry on C^1.

7.1 The ℓ^1 Quotient Norm

As established in Section 2, the scalar magnitude invariant under arbitrary local reference shifts is the ℓ^1 **quotient norm** on the space of cochains modulo exact gradients $C^1/\operatorname{im}(d_0)$:

$$\operatorname{Cost}(\delta) = \min_{f \in C^0} \|\delta - d_0 f\|_1$$

7.2 Descent to Cohomology

The gauge class $[\delta] = \delta + \operatorname{im}(d_0)$ is an affine subspace of $C^1(G)$. By linear programming duality, the minimum ℓ^1 norm on this affine space is attained by a representative intimately linked to the cycle structure of the graph. The optimization:

$$\operatorname{Cost}(\delta) = \min_{r \in [\delta]} \|r\|_1$$

depends only on the quotient/cohomology class $[\delta] \in H^1(G, \mathbb{R})$, abandoning the need for any ℓ^2 harmonic reference coordinate.

7.3 The Cohomological Norm

Proposition 7.1 (Existence of Minimizers). *The minimum* $\min_{f \in C^0} \|\delta - d_0 f\|_1$ *is attained for every* $\delta \in C^1(G)$.

Proof. The objective $f \mapsto \|\delta - d_0 f\|_1$ is continuous and coercive on $C^0/\ker(d_0) \cong \mathbb{R}^{|V|-1}$ (since $\ker(d_0) = \operatorname{span}(\mathbf{1})$ for connected G, the restriction $d_0|_{\operatorname{span}(\mathbf{1})^\perp}$ is injective, and by equivalence of norms on finite-dimensional spaces there exists $c > 0$ with $\|d_0 f\|_1 \geq c\|f\|_2$ for all $f \perp \mathbf{1}$, hence $\|d_0 f\|_1 \to \infty$ as $\|f\|_2 \to \infty$ on this quotient). A continuous coercive function on a finite-dimensional space attains its infimum [9].
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Every minimizer $r^* = \delta - d_0 f^*$ *satisfies the first-order optimality condition* $B^T \phi^* = 0$ *for some* $\phi^* \in \partial \| \cdot \|_1(r^*)$ *[9, Section 23]. That is,* ϕ^* *is a divergence-free flow bounded entrywise by* 1.

Theorem 7.2 (Cohomological ℓ^1 Norm). *The functional*

$$\Phi : H^1(G, \mathbb{R}) \to \mathbb{R}_{\geq 0}, \qquad \Phi([\delta]) = \min_{r \in [\delta]} \|r\|_1$$

is a norm on $H^1(G, \mathbb{R})$.

Proof. **Positive definiteness.** If $\Phi([\delta]) = 0$, then by Proposition 7.1 the minimum is attained: there exists f^* with $\|\delta - d_0 f^*\|_1 = 0$, hence $\delta = d_0 f^*$, so $[\delta] = 0$. Conversely, $[\delta] = 0$ gives $\delta \in \mathrm{im}(d_0)$, so $\Phi([\delta]) = 0$.

Absolute homogeneity. For $\lambda \neq 0$: $\Phi([\lambda\delta]) = \min_f \|\lambda\delta - d_0 f\|_1 = \min_f \|\lambda(\delta - d_0(f/\lambda))\|_1 = |\lambda| \min_g \|\delta - d_0 g\|_1 = |\lambda|\Phi([\delta])$, where $g = f/\lambda$. For $\lambda = 0$: $\Phi([0]) = 0$.

Triangle inequality. Let f^*, g^* be minimizers for $[\delta]$ and $[\delta']$ respectively (existing by Proposition 7.1). Choosing $f = f^* + g^*$ in the minimization for $[\delta + \delta']$:

$$\Phi([\delta+\delta']) = \min_f \|\delta+\delta'-d_0 f\|_1 \leq \|(\delta-d_0 f^*)+(\delta'-d_0 g^*)\|_1 \leq \|\delta-d_0 f^*\|_1+\|\delta'-d_0 g^*\|_1 = \Phi([\delta])+\Phi([\delta'])$$

7.4 Sparsity Bias

Remark 7.3 (Geometric Sparsity Bias). ℓ^1 minimization inherently contacts faces of the crosspolytope geometry, often favoring lower-support representatives when the affine constraints permit them. Consequently, optimization may select representatives supported on smaller edge sets rather than expanding them as uniform fractional obstructions like the ℓ^2 harmonic projection inevitably does.

7.5 Structure on Cycle Bases and Minimum Weight Combinations

While minimizing ℓ^1 cost generally necessitates linear programming, the structural geometry of the cycle space allows for a direct evaluation on the graph's loops under favorable conditions.

Theorem 7.4 (Evaluation on Vertex-Disjoint Cycle Bases). *Suppose the non-tree part of G decomposes into pairwise vertex-disjoint simple cycles $\{C_1, \ldots, C_{\beta_1}\}$ (represented as signed indicator vectors for the cycles) attached by bridges, and the class $[\delta] = \sum_{i=1}^{\beta_1} a_i[C_i]$ has no residual contribution on bridges after exact reduction. Then the cohomological norm is:*

$$\Phi([\delta]) = \sum_{i=1}^{\beta_1} |a_i| \|C_i\|_1$$

Proof. Since the cycles are vertex-disjoint and bridge edges carry no residual after exact reduction, the potential $f \in C^0$ decomposes as $f = \sum_i f_i$ where f_i is supported on V_i. The exact form $d_0 f$ restricted to edge set E_i depends only on f_i (vertices outside V_i contribute zero to edges in E_i, and bridge edges are fully eliminated). Therefore:

$$\min_f \|\delta - d_0 f\|_1 = \sum_{i=1}^{\beta_1} \min_{f_i \in \mathbb{R}^{V_i}} \sum_{e \in E_i} |a_i (C_i)_e - (d_0 f_i)_e|$$

On each component, the cohomology class $[a_i C_i]$ is one-dimensional, and any exact correction on a single cycle changes the representative by $d_0 f_i$ for f_i supported on V_i. Since C_i is a single cycle, the minimum ℓ^1 representative of the class $[a_i C_i]$ has cost $|a_i| \|C_i\|_1$. On an oriented simple cycle, the signed sum of a representative around the cycle is invariant under adding $d_0 f$, since $\sum_{e \in C_i} (d_0 f)_e = 0$. Therefore every representative of $a_i C_i$ has ℓ^1-norm at least $|\sum_{e \in C_i} a_i (C_i)_e| = |a_i| \|C_i\|_1$, with equality achieved by the constant representative $a_i C_i$.

Remark 7.5 (Existence of Vertex-Disjoint Cycle Bases). A vertex-disjoint cycle basis exists precisely for *cactus graphs* (connected graphs in which every 2-edge-connected component is a single cycle). For general graphs—including K_4, whose $\beta_1 = 3$ independent cycles necessarily share vertices—no vertex-disjoint cycle basis exists, and the general case requires the linear program to resolve interactions, corresponding to the **minimum weight homologous chain** [12]. When cycles share vertices but not edges, the decomposition may still hold, but the proof requires a more delicate LP analysis showing that the shared-vertex coupling constraints are independently satisfiable.

Remark 7.6 (Geometry of the Cycle Polytope). Under the Φ functional, the metric geometry of $H^1(G)$ behaves structurally as a polyhedral metric space whose unit locus mirrors the network's bounded cycle polytope $(\mathrm{conv}\{\pm C_1, \ldots, \pm C_{\beta_1}\})$.

Remark 7.7 (Uniqueness of Φ). The norm Φ is unique in the following narrow sense: it is the norm obtained by the standard quotient construction from the fixed ℓ^1 norm on C^1 through the quotient map $C^1 \to C^1/\operatorname{im}(d_0)$. The construction is

$$\|[x]\|_{\mathrm{quot}} = \inf\{\|y\|_1 : y \in [x]\}.$$

It is not the unique possible norm on $H^1(G)$ or on $C^1/\operatorname{im}(d_0)$. Weighted ℓ^1 norms, ℓ^2 norms on harmonic representatives, and other Banach-space-induced norms are legitimate alternatives once the cochain geometry is changed. The claim is only that, given the ℓ^1 geometry on cochains, the standard quotient construction determines this Φ.

---

8. Total Variation Denoising and Fast Algorithms (Layer C)

The gauge optimization $\min_f \|\delta - Bf\|_1$ exhibits a structural relationship to anisotropic total-variation and incidence-regression formulations extensively developed in image filtering [2, 13]. It is not identical to every standard TV denoising formulation, such as ROF denoising with an ℓ^2 data term. Its role here is different: it arises as the quotient norm induced by the assumed ℓ^1 geometry on coboundary spaces, rather than as a regularization principle or signal recovery objective. Setting the mapping between computational models:

Imaging	Defect Topology
Image Signal	Vertex Potential (f)
Image Gradient (∇)	Exterior Derivative ($d_0 = B$)
Observed Gradient (y)	Source Defect Field (δ)
TV Residual ($y - \nabla u$)	Topological Obstruction (r)

Consequently, the optimization objective directly mirrors **graph total variation (graph-TV)**. Where classical TV identifies structural borders (sparse semantic image edges), the framework can help identify quotient-residual structure associated with graph cycles.

Computational Remark. This overlap suggests that algorithms from TV and saddle-point convex optimization may be applicable after reformulation. Rather than utilizing generic $O(|E|^3)$ interior point bounds, the ℓ^1 quotient norm is computable using splitting protocols (Chambolle–Pock primal-dual [2], Split Bregman, ADMM), which admit per-iteration complexity scaling linearly in $|E|$. Whether these methods achieve comparable convergence rates in the graph cohomology setting is an empirical question not addressed here; we note only the formal structural correspondence.

---

9. Cut Duality and Flow Circulation (Layer B)

We establish that the topological magnitude of the norm is strictly characterized by valid internal network configurations.

Theorem 9.1 (Cut Duality).

$$\Phi([\delta]) = \max_{\phi} \{\langle \phi, \delta \rangle : \|\phi\|_\infty \leq 1,\ d_0^* \phi = 0\}$$

Proof. The primal problem $\min_f \|\delta - Bf\|_1$ is reformulated as a linear program by introducing auxiliary variables $t_e \geq 0$:

$$\text{Primal: } \min_{f,t} \sum_e t_e \quad \text{subject to} \quad -t_e \leq \delta_e - (Bf)_e \leq t_e, \quad t_e \geq 0$$

Let $\phi_e^+, \phi_e^- \geq 0$ be the dual variables for the upper and lower bound constraints respectively. Setting $\phi_e = \phi_e^+ - \phi_e^-$, the dual problem becomes:

$$\text{Dual: } \max_{\phi} \langle \phi, \delta \rangle \quad \text{subject to} \quad B^T \phi = 0, \quad |\phi_e| \leq 1 \text{ for all } e$$

The constraint $B^T \phi = 0$ requires ϕ to be divergence-free (a circulation). The notation $d_0^* \phi = 0$ is used only when the standard ℓ^2 pairing is being invoked;

the LP dual constraint itself is the purely algebraic transpose equation $B^T\phi = 0$. The bound $|\phi_e| \leq 1$ constrains $\phi \in B_\infty$. Strong duality holds because both the primal and dual are feasible finite-dimensional LPs: the primal is feasible ($f = 0$ is feasible) with objective bounded below by zero, and the dual is feasible ($\phi = 0$ satisfies $B^T\phi = 0$ and $\|\phi\|_\infty = 0 \leq 1$) [4, Chapter 5].

Remark 9.2 (Dual Norm of Bounded Circulations). The transpose-divergence condition $B^T\phi = 0$ fixes $\phi \in Z_1(G)$, the cycle space. The bounded infinity constraint yields:

$$\Phi([\delta]) = \max_{\phi \in Z_1(G) \cap B_\infty} \langle \phi, \delta \rangle$$

Hence, Φ is represented by the support function of the bounded circulation polytope $Z_1(G) \cap B_\infty$, equivalently the dual formula induced by the quotient norm. Within the network interpretation, it isolates the maximal potential flow pairing compatible with the measured contradiction field δ. In contrast to ℓ^2 Hodge theory, which decomposes via orthogonal projection, the ℓ^1 cohomological structure is governed by convex duality between defect fields and bounded circulations.

---

10. The ℓ^1-Native Decomposition (Layer B)

Given the ℓ^1 geometry on the coboundary space, the quotient-minimizing decomposition is not orthogonal, but derived from primal-dual optimality. It provides the canonical structure for measuring cohomological magnitude under ℓ^1 geometry, whereas ℓ^2 Hodge decomposition provides an orthogonal coordinate system.

Definition 10.1 (Primal Minimizer). The primal minimization problem $\min_{f \in C^0} \|\delta - d_0 f\|_1$ yields an optimal vertex potential f^*.

Definition 10.2 (Dual Witness). The dual maximization problem from Theorem 9.1 yields an optimal bounded circulation $\phi^* \in Z_1(G) \cap B_\infty$.

Remark 10.2a (ℓ^2-Independence of the Dual Constraint). The dual constraint is $B^T\phi = 0$, where B^T is the transpose of the incidence matrix. Calling this map d_0^* is justified only after choosing the standard ℓ^2 pairing. The primal-dual framework of this section uses only the incidence matrix B, its transpose B^T, the ℓ^1 norm on C^1, and convex optimization.

Theorem 10.3 (An ℓ^1-Optimal Quotient Decomposition). *Every defect field $\delta \in C^1(G)$ admits an ℓ^1-optimal quotient decomposition:*

$$\delta = \delta_{\text{exact}} + \delta_{\text{irr}}$$

where: 1. $\delta_{\text{exact}} = d_0 f^*$ *for some primal minimizer* $f^* \in C^0$ *of* $\|\delta - d_0 f\|_1$. 2. $\delta_{\text{irr}} = \delta - d_0 f^*$ *is an ℓ^1-minimal representative of* $[\delta]$, *satisfying* $\|\delta_{\text{irr}}\|_1 = \Phi([\delta])$.

This optimal decomposition is certified by a dual witness ϕ^* satisfying:

$$d_0^*\phi^* = 0, \quad \|\phi^*\|_\infty \leq 1, \quad \langle \phi^*, \delta_{\text{irr}} \rangle = \|\delta_{\text{irr}}\|_1$$

The ℓ^1-minimal representative δ_{irr} may not be unique: the set of optimal representatives $\{r \in [\delta] : \|r\|_1 = \Phi([\delta])\}$ forms a convex polytope (as a level set of the convex objective on an affine subspace). However, the value $\Phi([\delta]) = \|\delta_{irr}\|_1$ is uniquely determined. The decomposition is generally not unique; only the quotient value $\Phi([\delta])$ is canonical.

Remark 10.4 (Sparsity vs. Diffusion). In ℓ^2, the splitting $\delta = d_0\alpha + h$ is a linear projection where h minimizes the sum of squares. In the ℓ^1-native decomposition, the splitting pairs a primal exact correction with a polyhedral dual flow. Because ℓ^1 minimization contacts faces of the crosspolytope, δ_{irr} may concentrate mass on smaller supports. The cycle-graph inflation ratio $(3 - 2/n)$ quantifies this mismatch only for the single-edge construction of Proposition 6.7.

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11. Conditional Extensions: Vector-Valued and Operator-Valued Defects (Future Work / Layer B-C)

The extension of quotient mapping to vector-valued fields formally branches according to the declared local symmetry monoid and observer class:

- **Coordinate-separable vector defects:** When the observer measures independent coordinates with no continuous gauge interaction, the scalar logic extends componentwise to the separable norm $\ell^1(E; \ell^1)$, yielding $\Phi([\delta]) = \sum_{j=1}^{k} \Phi([\delta^{(j)}])$.
- **Euclidean vector bond defects:** When local measurement exhibits continuous $\mathrm{SO}(k)$ rotational invariance, the mixed norm $\ell^1(E; \ell^2)$ is the corresponding observer geometry. The resulting quotient structure $\Phi_{2,1}([\delta]) = \inf_{\alpha \in C^0} \sum_e \|\delta_e - (d_0\alpha)_e\|_2$ measures rotational boundary offsets without coordinate aliasing.

Quantum operator-valued defects. A later paper will treat quantum operator-valued defect fields. The only claim made here is formal and finite-dimensional: if edge defects take values in a finite operator space and a trace-norm edge gauge has already been declared, then the same quotient-norm template can be written as

[Φ

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12. Discussion

12.1 What This Paper Establishes

Given an ℓ^1 norm at the cochain level, the cohomological measurement compatible with that geometry is the ℓ^1 quotient norm Φ, governed by convex duality rather than orthogonal decomposition. No ℓ^2 structure is needed to define Φ or its dual certificate. The ℓ^1-Native Decomposition provides the quotient-minimizing representative and a bounded circulation witness; the ℓ^2 Hodge decomposition remains an orthogonal coordinate system.

The conceptual progression is: 1. **Convex-Geometric Cohomology** (Sections 2-4): Establishes the topological framework. 2. ℓ^1/ℓ^2 **Misalignment** (Section 6, Layer B): Quantifies one explicit incompatibility between Euclidean projection and crosspolytope geometry, measuring ℓ^2 component-cost inflation at a ratio of $3 - 2/n$ for single-edge defects on cycle graphs. 3. **Cohomological norm** (Theorem 7.2, Layer B): Formalizes $\Phi([\delta]) = \min_r \|r\|_1$ as the ℓ^1 quotient norm evaluating the defect's cycle space structure. 4. **Primal-Dual Native Decomposition** (Section 10, Layer B): Elevates Cut Duality (Theorem 9.1) to define the ℓ^1 quotient-minimizing decomposition of a defect field into a removable potential and an irreducible residual, with a bounded circulation certificate. 5. **Image processing parallelism** (Section 8, Layer C): Notes the structural correspondence to TV denoising methods.

12.2 Relation to Prior Work

The discrete de Rham complex on graphs is due to Eckmann [1]. The bipartite Hodge decomposition on graphs without 2-cells is standard (see Lim [8] for a modern treatment). The LP duality formulation of Φ connects to classical network flow theory (Ford–Fulkerson [3]) and the coarea formula for graph total variation (Chambolle et al. [2]). The ℓ^1 quotient norm on cohomology is related to Gromov's bounded cohomology [11], though the finite-dimensional setting here avoids the subtleties of the infinite-dimensional theory. The quotient norm construction itself is standard in functional analysis [9]. The discrete calculus framework is developed systematically in [10].

12.3 Position in the Unified Program

This paper resolves a narrower structural question: given a declared ℓ^1 geometry on coboundary defects, what quotient norm, decomposition, and dual certificate does that choice induce?

The logical dependency stack is:

1. **Observer discipline (Paper 000)** [18]: Quotient obstruction, observer passivity, contextual congruence, and licensed lift boundaries.
2. **Gauge-classification level** [15, 16]: The declared observer assumptions select an ℓ^1 cochain geometry in the relevant scalar edge-additive regime.

3. **Dynamical source level** [14]: Projection of linear dynamics onto non-linear manifolds generates intrinsic coboundary defects measured in this geometry.
4. **Cohomological level (this paper):** The declared ℓ^1 geometry induces a quotient norm Φ on $C^1/\operatorname{im}(d_0)$, hence on H^1 for graphs without 2-cells. This is the norm obtained by the standard quotient construction from the ℓ^1 structure on C^1 (Remark 7.7). The ℓ^2 Hodge decomposition provides a linear parameterization of cohomology classes; Φ measures their ℓ^1 quotient magnitude.

Thus, within this declared observer class, the measurement, decomposition, and optimization of defect fields are not independent choices; they are linked by the selected cochain geometry and the standard quotient construction. Outside that observer class, different local gauges produce different quotient norms.

12.4 Connection to Observer Theory and Contextual Congruence

The quotient norm Φ developed here is the finite graph instance of the General Theory normed cokernel [18]. In that framework, observers are repair-compatible quotient maps satisfying $\Phi_{\mathcal{O}}(\mathcal{O}(q)) \leq \Phi(q)$ (observer passivity, Theorem O1 in [18]). The Φ computed by ℓ^1 quotient minimization is the diagnostic that observers process; passivity constrains how much of the certified obstruction can be erased.

The lift trigger in the general framework is context-visible non-congruence: a witness $w = (x, y, c)$ with $p(x) = p(y)$ but $p(\rho(c)x) \neq p(\rho(c)y)$. A positive quotient norm alone does not license a lift. When $\Phi([\delta]) > 0$ is exposed by a declared context witness, a lift may be licensed only through the corresponding finite lift contract. The primal-dual certificate (f^*, ϕ^*) from Theorem 10.3 provides numerical evidence for non-exactness; it does not by itself certify semantic truth or lift authority.

Sheaf Gluing Obstruction The same quotient-norm construction applies to the finite sheaf gluing obstruction (Theorem L3 in [18]). Given a finite site $(\mathcal{C}, J)$, a finite presheaf F, and a finite declared cover $\mathcal{U}$, a compatible local family $a \in \ker(d_1)$ that does not glue to a global section defines a nonzero quotient obstruction $q_a \in H^1$. A declared quotient norm measures the cost of this gluing failure; in the graph ℓ^1 specialization, the bounded circulation witness certifies it.

For graphs without 2-cells (the setting of this paper), $\ker(d_1)$ is all of C^1, and the quotient obstruction reduces to the construction developed here. When 2-cells are present, the closure map $\overline{d}_1 : Q^1 \to C^2$ and the closure defect $\Gamma_2(q) = \|\overline{d}_1(q)\|$ provide the additional diagnostic layer classifying obstruction classes as exact, survivor, or open.

12.5 Limitations and Future Work

The ℓ^1 norm is assumed from the declared observer class and is not re-derived here. The inflation analysis in Section 6 is restricted to cycle graphs C_n with a single-edge localized defect; no universal inflation bound is claimed. The TV denoising analogy in Section 8 is structural rather than a convergence or performance theorem. The restriction to graphs without 2-cells means the coexact term vanishes and every nonzero quotient class is closed; extending the analysis to simplicial 2-complexes, where the full tripartite Hodge decomposition $\delta = d\alpha + \delta\beta + h$ and the closure map $\overline{d}_1$ are active, is a separate problem.

The quantum/operator-valued paragraph in Section 11 is a scoped future-work bridge only; no quantum admissibility, CPTP, or operator-duality theorem is proved here.

12.6 Closing Remark

The results of this paper do not introduce new optimization problems or algorithms; rather, they identify the structural role of an existing ℓ^1 minimization within a cohomological framework. When an observer has already selected ℓ^1 geometry on cochains, the measurement, decomposition, and dual certification of defect fields are constrained by the standard quotient norm and LP duality. The ℓ^1 quotient norm Φ links local defect aggregation, global topology, and convex optimization in a common framework.

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Appendix A: Cut Decomposition and ℓ^1 Duality on Finite Graphs

This appendix records the finite graph cut/coarea perspective supporting the ℓ^1 quotient geometry used in the main text.

A.1 Level Sets and Graph Cuts

The total variation (the discrete ℓ^1 gradient norm) connects directly to the coboundary operator:

$$\mathrm{TV}_G(v) = \|d_0 v\|_1 = \sum_{(x,y)=e\in E} |v_x - v_y|$$

We parameterize the scalar field into level sets via a threshold $\tau \in \mathbb{R}$:

$$V_\tau = \{x \in V \mid v_x > \tau\}$$

The boundary of any level set V_τ is a graph cut — the set of edges with exactly one endpoint in V_τ.

A.2 The Discrete Coarea Formula

Theorem A.1 (Discrete Coarea Formula). *For any $v \in \mathbb{R}^V$ on a graph G:*

$$\mathrm{TV}_G(v) = \int_{-\infty}^{\infty} |\partial V_\tau| \, d\tau$$

Proof. For a single edge $e = (x, y)$ with $v_x > v_y$:

$$|v_x - v_y| = \int_{-\infty}^{\infty} \mathbf{1}_{\tau\in[v_y,v_x)} \, d\tau = \int_{-\infty}^{\infty} \mathbf{1}_{e\in\partial V_\tau} \, d\tau$$

Summing over all edges and exchanging the (finite) sum and integral:

$$\mathrm{TV}_G(v) = \int_{-\infty}^{\infty} |\partial V_\tau| \, d\tau$$

Corollary A.2 (Discrete Summation Form). *If v takes r distinct values $v_{(1)} < v_{(2)} < \cdots < v_{(r)}$, then:*

$$\mathrm{TV}_G(v) = \sum_{j=1}^{r-1} (v_{(j+1)} - v_{(j)}) \, |\partial V_{v_{(j)}}|$$

A.3 The ℓ^1 Specificity

Proposition A.3 (ℓ^1 Specificity of the Linear Coarea Decomposition). *For $p > 1$, there exists no function $g_p : \mathbb{R} \to \mathbb{R}_{\geq 0}$ such that*

$$|a - b|^p = \int_{-\infty}^{\infty} g_p(\tau) \, \mathbf{1}_{\tau\in[\min(a,b),\max(a,b))} \, d\tau$$

for all $a, b \in \mathbb{R}$.

Proof. Setting $a = L > 0$ and $b = 0$ gives $L^p = \int_0^L g_p(\tau) \, d\tau$ for all $L > 0$. By the Fundamental Theorem of Calculus, $g_p(\tau) = p\tau^{p-1}$ a.e. on $(0, \infty)$. For general $a > b \geq 0$: $\int_b^a p\tau^{p-1} \, d\tau = a^p - b^p \neq (a - b)^p$ when $b > 0$ and $p > 1$. Contradiction.

A.4 TV-Cut Relaxation Tightness and Max-Flow Duality

Theorem A.4 (TV-Cut Relaxation Tightness). *For distinguished vertices $s, t \in V$ and $\mathcal{F}_{s,t} = \{v : V \to [0,1] \mid v_s = 1, v_t = 0\}$:*

$$\min_{v \in \mathcal{F}_{s,t}} \mathrm{TV}_G(v) = \min_{\substack{S \subseteq V \\ s \in S,\, t \notin S}} |\partial S|$$

Proof. Binary indicators $v = \mathbf{1}_S$ achieve $\mathrm{TV}_G(\mathbf{1}_S) = |\partial S|$, giving $\leq$. For $\geq$: by the coarea formula, $\mathrm{TV}_G(v) = \int_0^1 |\partial V_\tau| \, d\tau$, and each V_τ is a valid s-t cut.

By the Ford-Fulkerson max-flow/min-cut theorem:

$$\min_{v \in \mathcal{F}_{s,t}} \mathrm{TV}_G(v) = \min_S |\partial S| = \max_{f \in \mathrm{Flow}(s,t)} \mathrm{val}(f)$$

A.5 Relation to the Cohomological Norm

The coarea formula applies directly to exact edge fields of the form (d_0v), because such fields arise from thresholding a vertex potential (v). The cohomological quotient problem

[$\Phi([\delta]) = \min_\{f \in C^0\} |\delta - d_0 f|_1$]

is different: the residual $(\delta - d_0 f) is an arbitrary edge cochain in the affine class ([\delta]), not necessarily the gradient o$

A direct cut/coarea representation of $(\Phi) would require additional structure assigning level sets or cuts to arbitrary e$

Appendix A References

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Changelog

v5.1 (May 2026)

- Corrected Hodge-inflation language: the C_n calculation concerns ℓ^2 Hodge component bookkeeping cost, not a general failure of harmonic representatives to minimize the ℓ^1 quotient norm.

- Demoted vector/operator-valued material to conditional future-work bridge.
- Clarified that Appendix A gives coarea intuition for graph gradients, not a direct cut formula for arbitrary quotient residuals.
- Added General Theory-aligned status, scope, dependency, and non-claim metadata.
- Reframed Φ as the finite graph specialization of the normed cokernel $Q^1 = C^1/\operatorname{im}(d_0)$.
- Clarified that ℓ^1 is assumed from a declared observer class and is not re-derived here.
- Added the exact/survivor interpretation for graphs without 2-cells.
- Tightened observer-passivity and lift-authority language: positive Φ is diagnostic, not a licensed lift by itself.
- Updated stale internal paper titles and removed archived-source wording.

v5.0 (May 2026)

- Added §12.4: Connection to Observer Theory, Contextual Congruence, and Sheaf Gluing Obstruction.
- Cross-referenced Paper 000 v004 [18] (observer passivity, contextual congruence, reflective completion).
- Consolidated the General Theory connection into this paper.

v4.0 (April 2026)

- Audit-hardened pre-print with Layer A/B/C classification.